

Nihal Ahmed

PhD Candidate in Agricultural Economics and Management

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Research Profile

PhD researcher focusing on agricultural systems, ecological sustainability, and planetary boundaries. My research quantifies trade-embodied ecological pressures, with particular emphasis on ecosystem loss, land degradation, climate stress, and ecological equity. I combine multi-regional input-output modelling, spatial analysis, machine learning, and scenario modelling to investigate the links between consumption, trade, and ecological limits.

Education

PhD in Agricultural Economics and Management

Northwest A&F University, China

Supervisor: Prof. Huai Jianjun

Sep 2022 – Dec 2026

Master's in Applied Economics

Kohat University of Science and Technology, Pakistan

2019 – 2022

Research Experience

PhD Researcher

Northwest A&F University, China

Sep 2022 – Present

Trade-Embodied Ecosystem Accounting

- Developed consumption-based accounting frameworks to quantify belowground ecosystem multifunctionality (BEMF) embodied in agricultural trade.
- Integrated spatial soil-function data with agricultural production and trade data to trace ecosystem pressures from production regions to final consumption across 192 regions and 123 sectors.
- Quantified ecological pressures associated with China's agricultural imports and evaluated their distribution across exporting regions under climate change.

Deep Learning for Land-Use Scenario Synthesis

- Developed generative deep-learning models, including Variational Autoencoder (VAE), Conditional GAN (CGAN), and Recurrent Conditional GAN (RCGAN), to synthesize global land-use trajectories from CMIP6/LUH2 scenarios.
- Generated 30,000 physically plausible land-use pathways and developed a computationally efficient approach for exploring large scenario spaces.
- Combined Random Forest and SHAP analysis to identify major drivers of land degradation risk under SDG 15.3.1.

Provincial Ecological Efficiency and Spillover Networks

- Evaluated the efficiency of China's 31 provinces in transforming agricultural innovation into land ecological security using Slacks-Based Measure (SBM) efficiency models with undesirable outputs.

- Constructed interprovincial spillover networks using a modified gravity model and applied CONCOR block modelling and MRQAP to investigate structural and functional patterns of ecological spillovers.
- Identified factors associated with interprovincial differences and spillover relationships in agricultural ecological security.

Selected Publications

Belowground ecosystem multifunctionality embodied in China's agricultural imports: thermal geography, equity trajectory, and a multi-indicator accounting framework.

N. Ahmed, N.A. Khan, J. Ober, P. Sakiewicz, J. Huai.

Ecological Indicators, manuscript under review, 2026. **IF: 8.7** (JCR 2026)

Agricultural trade externalizes belowground ecosystem loss to climate-stressed exporting nations.

N. Ahmed, N.A. Khan, J. Ober, J. Huai.

International Journal of Sustainable Development & World Ecology, 2026. doi:10.1080/13504509.2026.2726387 **IF: 6.8** (JCR 2026)

Deep learning approaches for synthesizing Land Ecological Security pathways in global land-use scenarios.

N. Ahmed, N.A. Khan, J. Ober, M.F. Shahzad.

Resources, Environment and Sustainability, 2026. doi:10.1016/j.resenv.2026.100345 **IF: 9.2** (JCR 2026)

Efficiency divergence in agricultural innovation-to-ecological security transformation: Network analysis of Chinese provincial spillovers.

N. Ahmed, J. Ober, J. Huai.

Environmental and Sustainability Indicators, 2026. doi:10.1016/j.indic.2026.101224 **IF: 5.5** (JCR 2026)

Quantifying China's ecological assets: A carbon-based assessment of climate change and land use change effects using generalized additive models.

N. Ahmed, N.A. Khan, J. Ober, J. Huai.

Sustainable Futures, 2025. doi:10.1016/j.sfr.2025.101350 **IF: 6.7** (JCR 2026)

Synergistic evaluation of energy security and environmental sustainability in BRICS geo-political entities: An integrated index framework.

N. Ahmed, F. Ore Areche, J.M. More López, V.G. Sánchez Araujo, J.A. Legua Cárdenas, J. Ober.

Equilibrium. Quarterly Journal of Economics and Economic Policy, 2024. doi:10.24136/eq.3088 **IF: 6.7** (JCR 2026)

Publication record: 19 publications published or in press; 16 first-author publications.

Full publication list: [Google Scholar](#) | [nihalahmed.uk](#)

Research Methods & Technical Skills

Environmental & Trade Modelling

- Multi-Regional Input–Output (MRIO) analysis
- Consumption-based ecological accounting
- Structural Decomposition Analysis (SDA)
- Ecosystem multifunctionality assessment
- Agricultural and ecological footprint analysis

Machine Learning & Computational Modelling

- Variational Autoencoder (VAE)
- Conditional GAN (CGAN)
- Recurrent Conditional GAN (RCGAN)
- Random Forest
- XGBoost
- SHAP / Explainable AI

- Generalized Additive Models (GAM)
- Monte Carlo simulation

Efficiency, Econometrics & Network Analysis

- SBM efficiency analysis with undesirable outputs
- Simar–Wilson bootstrap
- Panel econometrics
- Two-Way Fixed Effects
- Spatial Durbin Models
- Panel Granger Causality
- Social Network Analysis
- Modified Gravity Models
- MRQAP

Software

R | Python | Stata | MATLAB | Adobe Illustrator

Environmental & Earth-System Data

CMIP6 | LUH2 | FABIO | SPAM | SoilGrids | WorldClim | ESA CCI | MODIS

Research Funding

National Natural Science Foundation of China (NSFC)

Grant No. 42075172

Principal Investigator: Prof. Huai Jianjun

Role: Graduate Research Team Member

Teaching Experience

Lecturer

Government College of Technology, Pakistan

2021 – 2022

Academic Service

Peer Reviewer

- Completed 115+ manuscript reviews for 22 international journals.
- Reviewer for journals including *Ecological Indicators*, *Agricultural Systems*, and *Journal of Environmental Management*.

References

Prof. dr hab. inż. Józef Ober

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Prof. Huai Jianjun

Professor; Doctoral Supervisor

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